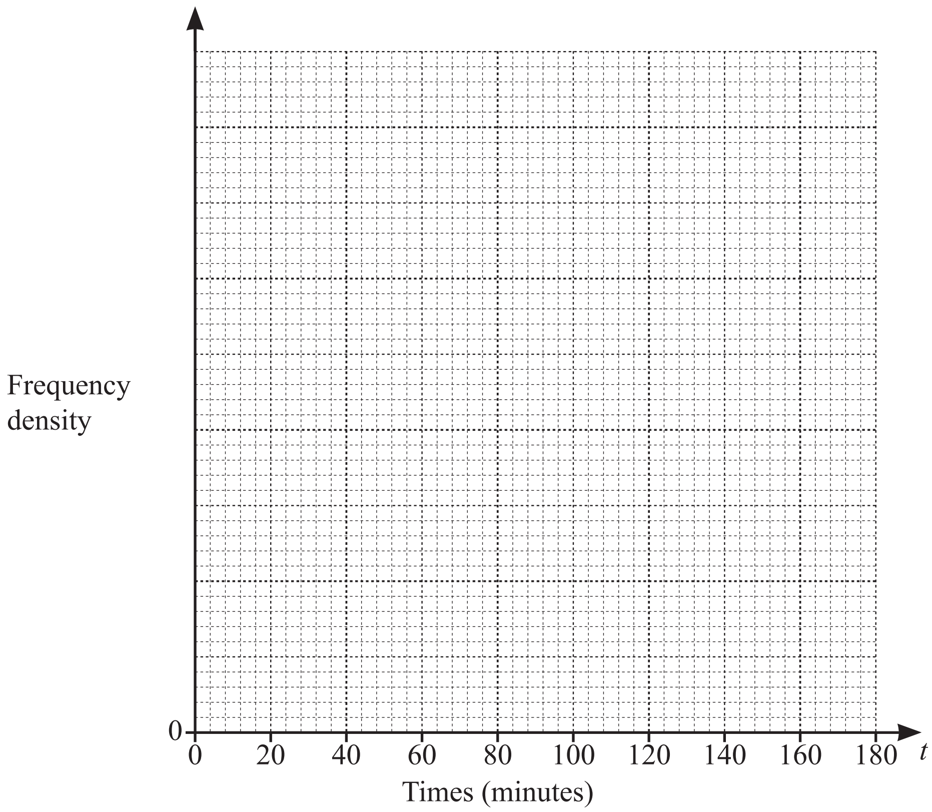
1 **(a)** One afternoon, there were 200 visitors to a library.
The table summarises the time, in minutes, each visitor spent in the library.

Time (t minutes)	$0 < t \leqslant 20$	$20 < t \leqslant 40$	$40 < t \leqslant 60$	$60 < t \leqslant 90$	$90 < t \leqslant 180$
Frequency	26	76	56	24	18

(i) On the grid, draw a histogram to represent this data.



[3]

(ii) Work out the percentage of these visitors who spent more than 40 minutes in the library.

.... % [2]

(b) Mario recorded the number of books borrowed by each of the 200 visitors to the library.
His results are shown in the table.

Number of books	0	1	2	3	4	5	6
Frequency	17	47	42	28	32	21	13

(i) Find the median.

.... [1]

(ii) Calculate the mean.

.... [2]

(iii) One of the visitors is selected at random.

Find the probability that this visitor borrowed more than 4 books.

.... [1]

(iv) Two of the visitors are selected at random.

Find the probability that only one of them borrowed 6 books.
Give your answer as a decimal correct to 4 significant figures.

.... [3]

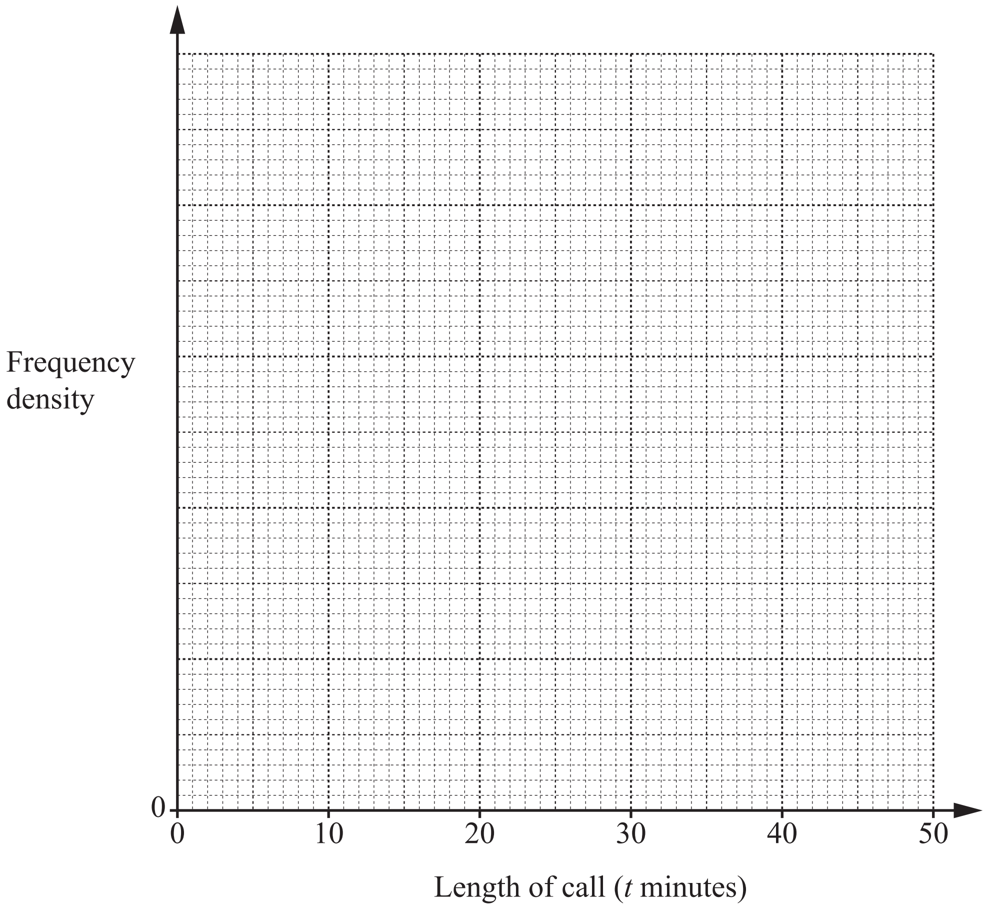
2 Sunil recorded the lengths, in minutes, of the 150 phone calls he made one month.
His results are summarised in the table.

Length of call (t minutes)	$0 < t \leqslant 5$	$5 < t \leqslant 10$	$10 < t \leqslant 20$	$20 < t \leqslant 30$	$30 < t \leqslant 50$
Frequency	35	42	30	28	15

(a) Calculate an estimate of the mean length of a call.

Answer .... minutes [3]

(b) On the grid below, draw a histogram to represent this data.



[3]

(c) Find an estimate for the percentage of Sunil's calls that were longer than 25 minutes.

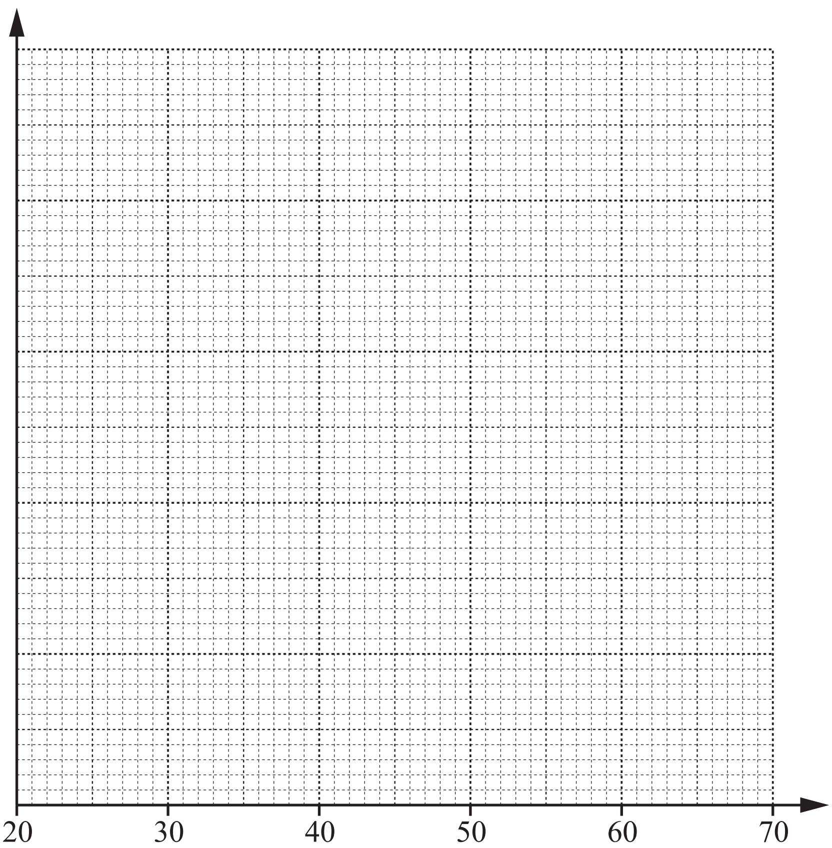
Answer .... % [2]

3 **(a)** The times taken by 135 runners to complete a cross-country course were recorded.
The results are summarised in the table.

Time (t minutes)	$20 < t \leqslant 30$	$30 < t \leqslant 35$	$35 < t \leqslant 40$	$40 < t \leqslant 50$	$50 < t \leqslant 70$
Number of runners	15	30	40	35	15

(i) On the grid, draw a histogram to represent this information.

Frequency density

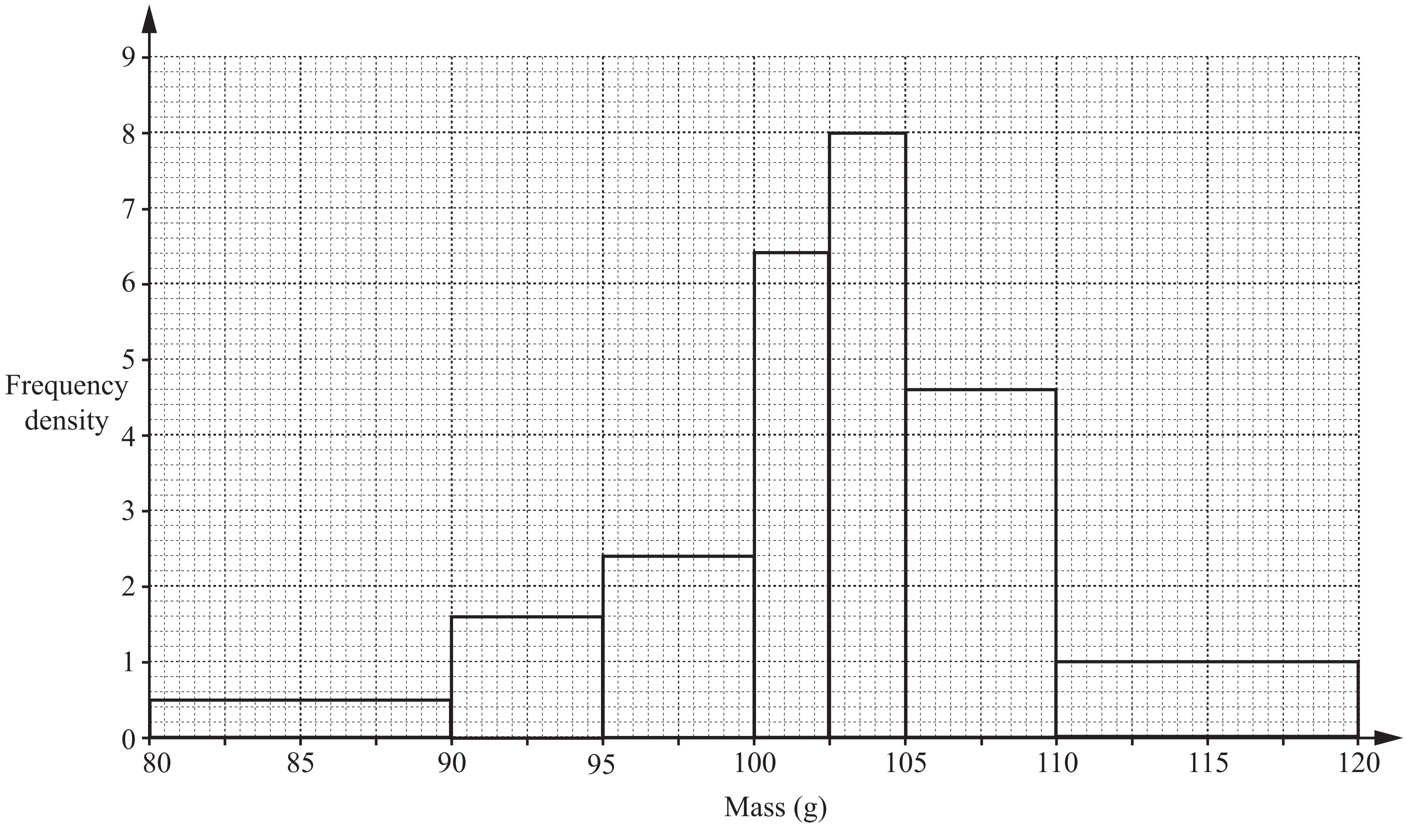


[3]

(ii) Calculate an estimate of the mean time.

Answer .... minutes [3]

4 **(a)** The histogram represents the distribution of the masses, in grams, of individual apples in a box.



This information is summarised in the table below.

Mass (m g)	Frequency
$80 < m \leqslant 90$	5
$90 < m \leqslant 95$	8
$95 < m \leqslant 100$	p
$100 < m \leqslant 102.5$	q
$102.5 < m \leqslant 105$	20
$105 < m \leqslant 110$	23
$110 < m \leqslant 120$	10

Calculate p and q.

Answer p = ... q = ... [2]

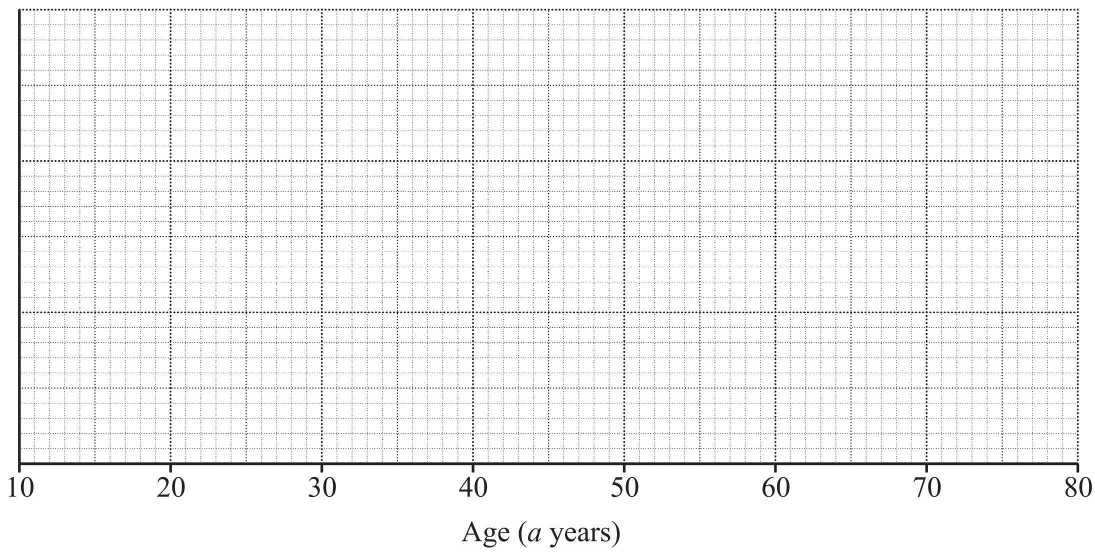
5 **(b)** The table summarises the ages of the members of a film club.

Age (a years)	$15 \leqslant a < 20$	$20 \leqslant a < 30$	$30 \leqslant a < 40$	$40 \leqslant a < 60$	$60 \leqslant a < 80$
Frequency	12	36	45	33	24

(i) Calculate an estimate of the mean age of the members.

Answer .... [3]

(ii) On the grid below, draw a histogram to represent this data.



[3]

(iii) Find an estimate for the number of members of the film club who are over 50.

Answer .... [1]

6 **(b)** The frequency table shows information about the mass of each of 50 trucks.

Mass (m kg)	$2000 < m \leqslant 2600$	$2600 < m \leqslant 3500$	$3500 < m \leqslant 5000$	$5000 < m \leqslant 5700$
Frequency	12	15	16	7

(i) Calculate an estimate for the mean mass of the trucks.

.... kg [4]

(ii) In a histogram showing this information, the height of the first block is 6 cm.

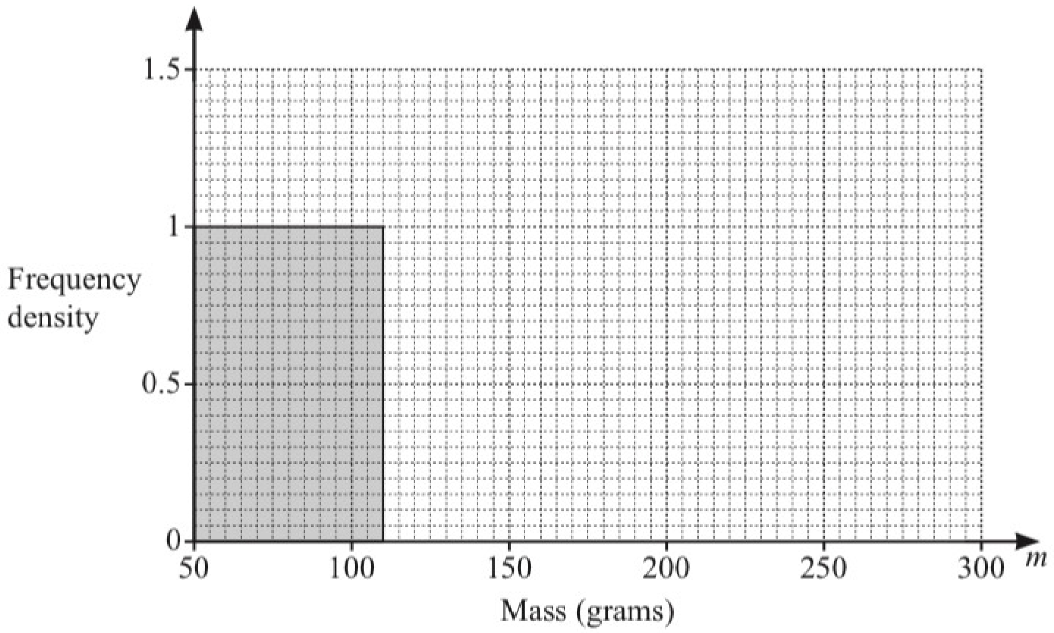
Calculate the heights of the remaining three blocks.

Height of block for $2600 < m \leqslant 3500$ .... cm

Height of block for $3500 < m \leqslant 5000$ .... cm

Height of block for $5000 < m \leqslant 5700$ .... cm [3]

7 Complete the histogram to show the information in the table.



[2]

8 A histogram is drawn showing the information in **Kai's** frequency table.
The height of the block for the interval $200 < d \leqslant 300$ is 2.6 cm.

Calculate the height of the block for each of the following intervals.

$80 < d \leqslant 100$ .... cm

$150 < d \leqslant 200$ .... cm

$300 < d \leqslant 400$ .... cm [3]

Kai's frequency table:

Distance (d km)	$80 < d \leqslant 100$	$100 < d \leqslant 150$	$150 < d \leqslant 200$	$200 < d \leqslant 300$	$300 < d \leqslant 400$
Frequency	7	33	76	52	32

9 150 students each record how far they can throw a tennis ball.
The table shows the results.

Distance (d metres)	$0 < d \leqslant 20$	$20 < d \leqslant 30$	$30 < d \leqslant 35$	$35 < d \leqslant 45$	$45 < d \leqslant 60$
Frequency	4	38	40	53	15

A histogram is drawn to show this information.
The height of the bar representing $30 < d \leqslant 35$ is 12 cm.

Calculate the height of each of the other bars.

Distance (d metres)	Frequency	Height of bar (cm)
$0 < d \leqslant 20$	4	
$20 < d \leqslant 30$	38	
$30 < d \leqslant 35$	40	12
$35 < d \leqslant 45$	53	
$45 < d \leqslant 60$	15	

[3]

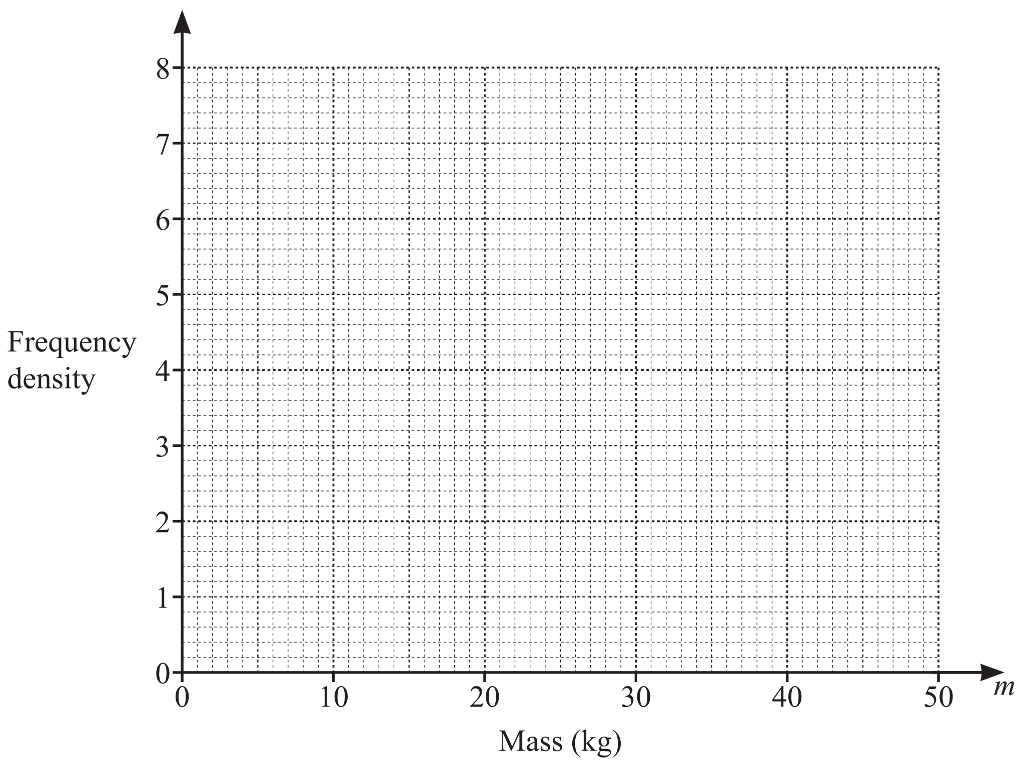
10 Information about the mass, m kg, of each of 150 children is recorded in the frequency table.

Mass (m kg)	$0 < m \leqslant 10$	$10 < m \leqslant 20$	$20 < m \leqslant 25$	$25 < m \leqslant 40$	$40 < m \leqslant 50$
Frequency	12	38	32	50	18

(a) Calculate an estimate of the mean mass.

.... kg [4]

(b) Draw a histogram to show the information in the table.



[4]

Answer Key:

1. W19/qp21/q2

2(a)(i)	Correct histogram with frequency density axis scaled	**3**	**B1** for 4 or more rectangles on correct bases **B1** for 4 or more correct frequency densities soi
2(a)(ii)	49	**2**	**M1** for 56 + 24 + 18 soi
2(b)(i)	2	**1**	
2(b)(ii)	2.63	**2**	**M1** for $([0 \times 17] + 1 \times 47 + 2 \times 42 + 3 \times 28 + 4 \times 32 + 5 \times 21 + 6 \times 13) \div 200$ oe
2(b)(iii)	$\frac{34}{200}$ oe	**1**	

2. W17/qp21/q2(b)

2(b)	Correct histogram with linear scale on frequency density axis	**3**	**B2** for all 5 bar heights correct with frequency density axis scaled OR **B1** for at least 3 correct heights drawn or 3 correct frequency densities calculated **B1** for 5 bars correct width and position

3. W16/qp21/q10(a)(i)

10 (a) (i)	Correct histogram with linear scale on frequency density axis	**3**	**B2** for all 5 heights correct with axis scaled OR **B1** for at least 3 correct frequency densities soi and **B1** for all 5 bars correct widths

4. W14/qp21/q4(a)

4 (a)	$p = 12, q = 16$	**2**	**B1** for one correct Or **M1** for $k \times 5$ or $l \times 2.5$ where k and l are attempts to read from the histogram

5. S14/qp22/q7(b)

(b) (i)	40.1	3	**M1** for $12 \times 17.5 + 36 \times 25 + 45 \times 35 + 33 \times 50 + 24 \times 70$ **M1** for division by *their* (12 + 36 + … + 24)
(ii)	Correct histogram	3	**B1** for 5 bars correct width and position **B1** for at least 3 correct heights $k \times (2.4, 3.6, 4.5, 1.65, 1.2)$ **B1** for 5 correct heights
(iii)	38 or 39 or 40 or 41	1	

6. S23/qp43/q6(b)(ii)(0580)

6(b)(ii)	5 3.2 3	3	**B1** for each If 0 scored, **SC1** for 3 frequency densities $\frac{12}{600}, \frac{15}{900}, \frac{16}{1500}, \frac{7}{700}$ seen oe to 3sf or better or multiplier 3 or 300

7. M23/qp42/q2(c)(ii) (0580)

2(c)(ii)	Correct histogram completed with widths 110 to 200 and 200 to 300 and heights 1.1 and 0.41	2	**B1** for one correct block If 0 scored, **SC1** for 1.1 and 0.41 seen

8. W22/qp42/q3(a)(iii) (0580)

3(a)(iii)	1.75 7.6 1.6	3	**B2** for two correct heights or **B1** for one correct height or 3 correct frequency densities or **M1** for scale factor of 5 or 0.2

9. W22/qp41/q5(b)(ii) (0580)

5(b)(ii)	0.3, 5.7, ..., 7.95, 1.5	3	**B2** for any two correct or **B1** for one correct or for at least three frequency densities seen 0.2, 3.8, 8, 5.3, 1 oe or **M1** for [factor] 1.5

10. S22/qp42/q7(a)(b) (0580)

7(a)	25.2 or 25.23…	4	**M1** for midpoints **soi** **M1** for use of $\sum fx$ with x in correct interval including both boundaries **M1 (dep** on 2nd M1) for $\sum fx \div 150$
7(b)	5 correct blocks	4	**B3** for 4 correct blocks or **B2** for 3 correct blocks or **B1** for 2 correct blocks or block widths 10, 10, 5, 15, 10 If 0 scored **SC1** for 4 correct frequency densities from 1.2, 3.8, 6.4, 3.33[3…] and 1.8 oe soi